

Short Intro to Decomposition Methods

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Preliminaries

- Brief introduction of the instructor
- Slides and lab(s) available on Github
- `https://github.com/benjisamschlu/course_decomposition_berkeley`
- Student introductions/familiarity with demography

Outline

- What is Demography?
- Aggregate measures
- Standardization and its limits
- Kitagawa decomposition
- Life tables
- Arriaga decompostion

What is Demography?

What is Demography?

"Demography is the scientific study of human population, including its size, distribution, composition, and the factors that determine changes in its size, distribution, and composition."(Shryock and Siegel 2004)

"Quantitative study of groups of people."(Lutz 2021)

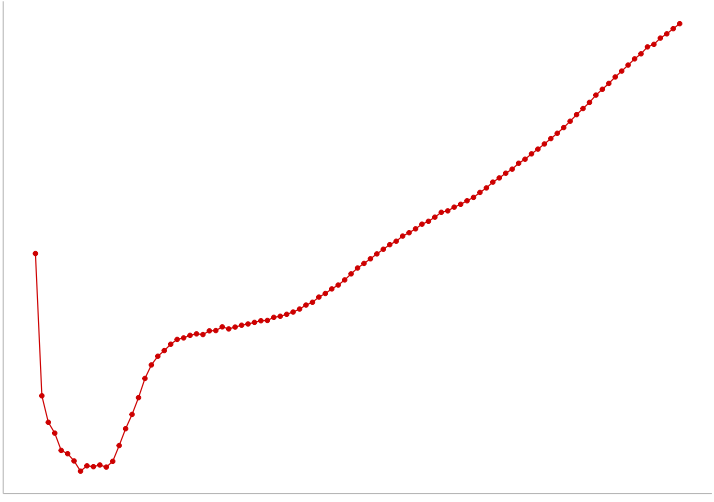
Demographic Balancing Equation

$$P(t+n) = P(t) + \ldots\ldots\ldots - \ldots\ldots\ldots + \ldots\ldots\ldots - \ldots\ldots\ldots \quad (1)$$

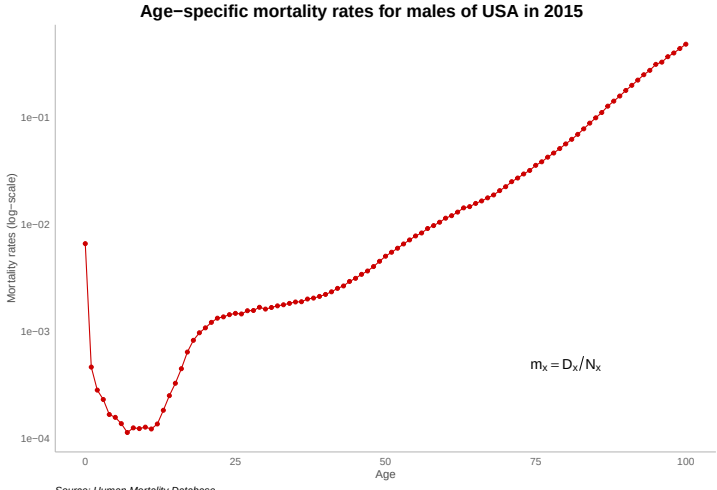
Demographic Balancing Equation

$$P(t+n) = P(t) + B[t, t+n] - D[t, t+n] + I[t, t+n] - O[t, t+n] \quad (2)$$

What are we looking at?

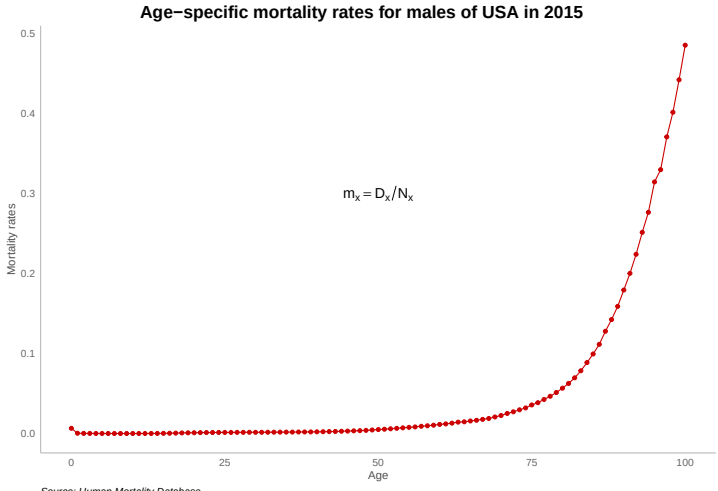


Mortality age schedule



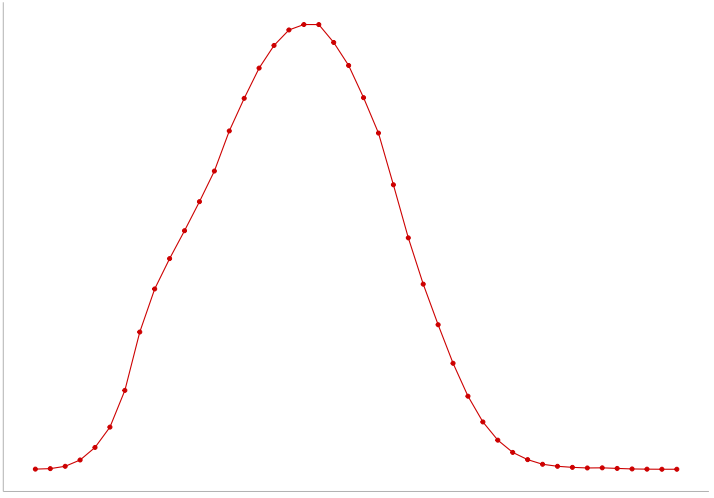
Source: Human Mortality Database

Mortality age schedule

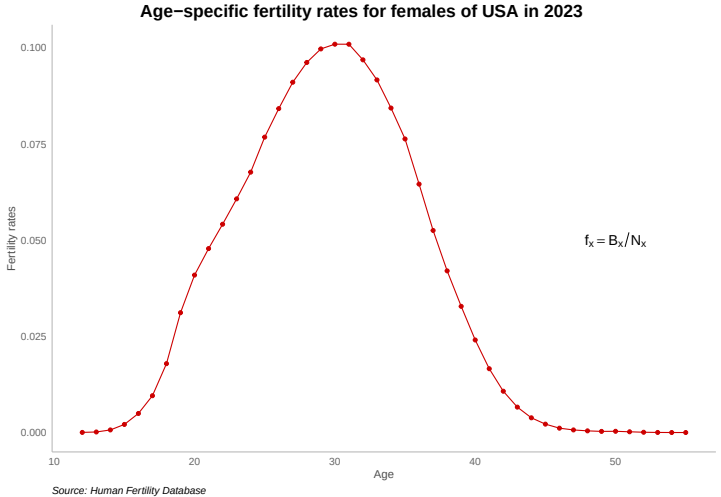


Source: Human Mortality Database

What are we looking at?



Fertility age schedule



Aggregate measures

- Summary measures of population change due to births, deaths or migration
  - Crude birth rates (?), general fertility rate (?), crude death rate (?), crude net migration rate (?), ...

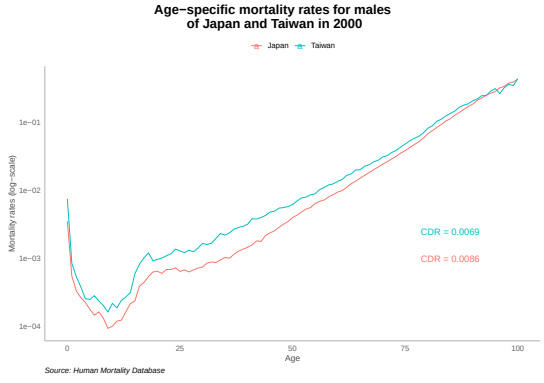
Aggregate measures

- Summary measures of population change due to births, deaths or migration
  - Crude birth rates (Bth/N), general fertility rate (Bth/N^f_{15-49}), crude death rate (Dth/N), crude net migration rate (I-O/N), ...

Aggregate measures

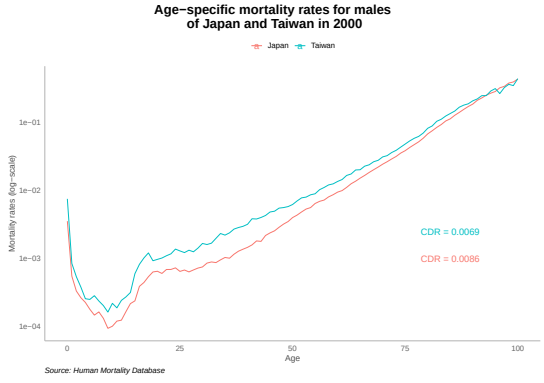
- Summary measures of population change due to births, deaths or migration
  - Crude birth rates (Bth/N), general fertility rate (Bth/N^f_{15-49}), crude death rate (Dth/N), crude net migration rate (I-O/N), ...
- We compare these measures across populations and over time

Crude death rate



Anything surprising?

Crude death rate



Anything surprising?

- **Rate effects** and **compositional effects**

Re-expressing crude death rate

$$CDR = \frac{D}{N} = \frac{\sum_x ?}{N} \tag{3}$$

(5)

(7)

(8)

Re-expressing crude death rate

$$CDR = \frac{D}{N} = \frac{\sum_x ?}{N} \tag{3}$$

$$= \frac{\sum_x M_x N_x}{N} \tag{4}$$

$$= \sum_x M_x \frac{N_x}{N} \tag{5}$$

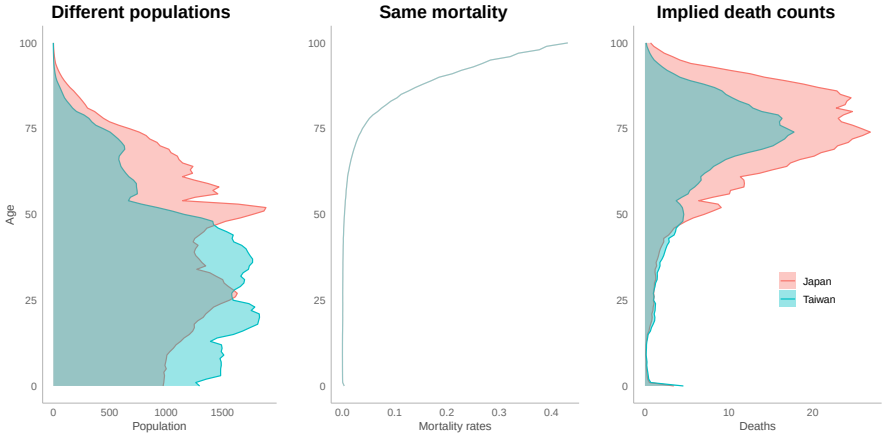
(7)

(8)

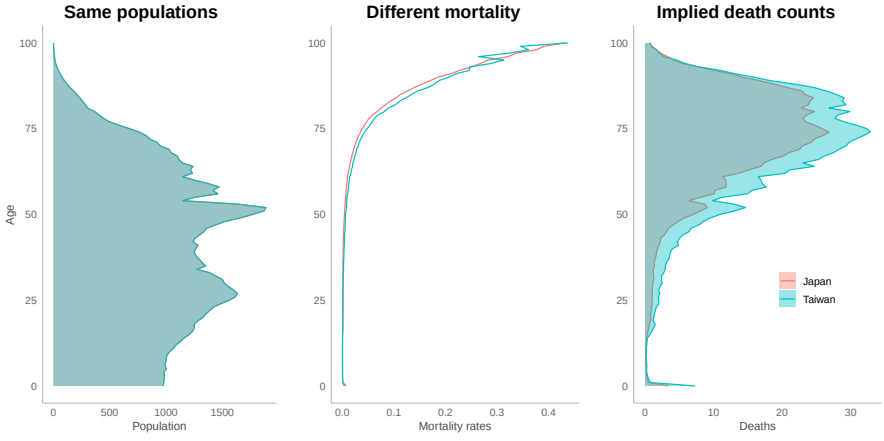
Re-expressing crude death rate

$$
\begin{aligned}
CDR = \frac{D}{N} &= \frac{\sum_x ?}{N} && (3)\\
&= \frac{\sum_x M_x N_x}{N} && (4)\\
&= \sum_x M_x \frac{N_x}{N} && (5)\\
&= \sum_x \underbrace{M_x}_{\text{Rates}} \underbrace{C_x}_{\text{Comp.}} && (6)\\
&= \sum_x w_x M_x && (7)\\
& && (8)
\end{aligned}
$$

Compositional effects



Rate effects



Difference between crude rates

$$\Delta CDR = CDR^{\text{Twn}} - CDR^{\text{Jpn}} \tag{9}$$

$$= \sum_x M_x^{\text{Twn}} C_x^{\text{Twn}} - \sum_x M_x^{\text{Jpn}} C_x^{\text{Jpn}} \tag{10}$$

$$= f(\Delta\text{rates}, \Delta\text{composition}) \tag{11}$$

Standardization methods

Direct standardization: Remove compositional effect and focus on the difference in rates.

How?

Standardization methods

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How? Compare CDR^{Jpn} with

$$CDR_{\text{Std}}^{\text{Twn}} = \sum_{x} M_x^{\text{Twn}} C_x^{\text{Jpn}} \tag{12}$$

Standardization methods

Direct standardization: Remove compositional effect and focus on the difference in rates.

How? Compare CDR^{Jpn} with

$$CDR_{\text{Std}}^{\text{Twn}} = \sum_x M_x^{\text{Twn}} C_x^{\text{Jpn}} \tag{12}$$

Indirect standardization:

$$SMR^{\text{Twn}} = \frac{\sum_x M_x^{\text{Twn}} C_x^{\text{Twn}}}{\sum_x M_x^{\text{Jpn}} C_x^{\text{Twn}}} = \frac{D^{\text{Twn}}}{\sum_x M_x^{\text{Jpn}} C_x^{\text{Twn}}} = \frac{D^{\text{Twn}}}{E^{\text{Twn}}} \tag{13}$$

Standardization: a partial solution

Limitations?

Standardization: a partial solution

Limitations?

- Choice of standard
- Dependence on standard
- Components contribution

Decomposition: improve comparison methods

What drives the difference in aggregate measures?

- Rates vs composition effect
- Contribution of each effect to difference in crude rates

Kitagawa decomposition

Kitagawa 1955 (see Appendix):

$$\Delta CDR = \underbrace{\sum_x \left(\frac{M_x^{Twn} + M_x^{Jpn}}{2} \right) \left(\frac{N_x^{Twn}}{N^{Twn}} - \frac{N_x^{Jpn}}{N^{Jpn}} \right)}_{\text{Difference due to pop. composition}} + \underbrace{\sum_x \left(\frac{\frac{N_x^{Twn}}{N^{Twn}} + \frac{N_x^{Jpn}}{N^{Jpn}}}{2} \right) \left(M_x^{Twn} - M_x^{Jpn} \right)}_{\text{Difference due to rates}} \quad (14)$$

Kitagawa decomposition

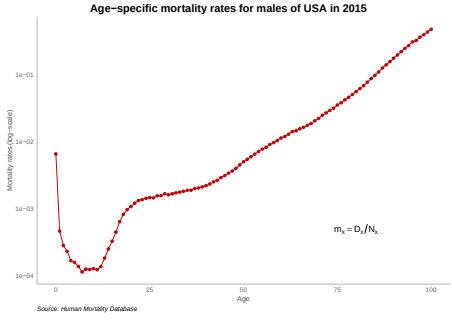
- **Composition effect**: difference in age composition, holding rates at the average of the two populations
  - How much of the difference in CDR is due to the population being younger of older?
- **Rate effect**: difference in mortality rates, holding the age composition at the average of the two populations
  - How much of the difference in CDR is due to the mortality rates being different?
- The effects can be greater or smaller than 0. The total difference in CDR is the sum of the two effects.

Life tables - machinery

Main input?

Life tables - machinery

Main input?



Convert ${}_nm_x$ into ${}_nq_x$ (${}_na_x$!)

Life tables - summary measures

Who is the star of the life table?

Life tables - summary measures

Who is the star of the life table?

- Life expectancy

Life tables - summary measures

Who is the star of the life table?

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Expectation matters but what about the spread?

Life tables - summary measures

Who is the star of the life table?

- Life expectancy

Expectation matters but what about the spread?

- Lifespan disparity

Life tables - cohort vs period

- What is the difference (think about the main input)?

Life tables - cohort vs period

- What is the difference (think about the main input)?
- Why do we use period life tables?

Life tables - cohort vs period

- What is the difference (think about the main input)?
- Why do we use period life tables?
  - Synthetic cohort
- Impact of Covid19: excess mortality

Arriaga decomposition

Arriaga 1984:

- Comparing life expectancies across populations and over time
- Contribution of each age group in life expectancy difference
  - Age group dependence: direct vs indirect effect

Lab

We will use data from the Human Mortality Database (HMD) and the Human Fertility Database (HFD).

Section A

2000 vs 2020, Japan females. Compute

1. CDR
2. SMR
3. Kitagawa decomposition for CDR

Section B

Chile vs Ireland females, 2020. Compute

1. GFR
2. Adapt Kitagawa decomposition for GFR

References I

- Arriaga, Eduardo E (1984). "Measuring and explaining the change in life expectancies". In: *Demography* 21.1, pp. 83–96.
- Kitagawa, Evelyn M (1955). "Components of a difference between two rates". In: *Journal of the american statistical association* 50.272, pp. 1168–1194.
- Lutz, Wolfgang (2021). *Advanced introduction to demography*. Edward Elgar Publishing.
- Shryock, Henry S. and Jacob S. Siegel (2004). *The Methods and Materials of Demography*. Ed. by Jacob S. Siegel and David A. Swanson. 2nd ed. San Diego, CA and London: Elsevier Academic Press.

Appendix - Kitagawa (1955)

$$\Delta CDR = \underbrace{\sum_x M_x^{Jpn}\left[\frac{N_x^{Twn}}{N^{Twn}} - \frac{N_x^{Jpn}}{N^{Jpn}}\right]}_{\text{Change in } x\text{-composition}} + \underbrace{\sum_x \frac{N_x^{Jpn}}{N^{Jpn}}\left[M_x^{Twn} - M_x^{Jpn}\right]}_{\text{Change in rates using pop 1 as the standard}} + \underbrace{\sum_x \left(M_x^{Twn} - M_x^{Jpn}\right)\left[\frac{N_x^{Twn}}{N^{Twn}} - \frac{N_x^{Jpn}}{N^{Jpn}}\right]}_{\text{Interaction between rates and composition}} \quad (15)$$